

# ENEEL KUMAR A

+91-9606288164 | veneeldas77@gmail.com | linkedin.com/in/veneelkumara77 | github.com/Veneel77 | Bangalore, India

## SUMMARY

AI Engineer specialising in computer vision, deep learning, and MLOps — designing, training, and deploying CNN-based imaging AI models and production-grade AI pipelines using Python, TensorFlow, FastAPI, Docker, and AWS. Peer-reviewed research publication in medical imaging AI: CNN transfer learning (VGG16, ResNet50) on MRI datasets achieving >90% accuracy. Experienced in cloud-native deployment, CI/CD, observability, and end-to-end MLOps with strong foundations in data structures, algorithms, and agile software engineering.

## SKILLS

|                                 |                                                                                                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LLM &amp; Agentic AI:</b>    | LLM-Powered Agents, LLM APIs (Gemini, HuggingFace, OpenAI-compatible), LangChain, CrewAI, Prompt Engineering (System Prompts, Few-Shot, Structured Outputs), Tool-Calling Interfaces |
| <b>RAG &amp; Vector Search:</b> | RAG Architecture, Document Ingestion, Chunking Strategies, chatbot , Embedding Generation, Vector Search, ChromaDB, Semantic Retrieval, Agent Evaluation (RAGAS concepts)            |
| <b>ML &amp; Deep Learning:</b>  | TensorFlow, Keras, Scikit-Learn, XGBoost, HuggingFace, CNN, Transfer Learning, Model Evaluation, Hyperparameter Tuning                                                               |
| <b>Backend &amp; APIs:</b>      | FastAPI, REST APIs, Microservices Architecture, API Design, Tool-Calling Integration, CI/CD, Docker, Docker Compose, Kubernetes                                                      |
| <b>Cloud Platforms:</b>         | AWS (SageMaker, EC2, S3, IAM), GCP                                                                                                                                                   |
| <b>Programming Languages:</b>   | Python, Java, SQL                                                                                                                                                                    |
| <b>Software Engineering:</b>    | OOP, Data Structures & Algorithms, SDLC, Code Review, Git-Based Workflows, Version Control                                                                                           |
| <b>Data Engineering:</b>        | Airflow, ETL/ELT Pipelines, Pandas, NumPy, SQL                                                                                                                                       |

## EXPERIENCE

### AI/ML Engineer Trainee

Sep 2024 – Feb 2025

Rooman Technologies · Remote · 6 months

- Engineered and optimised end-to-end ML pipelines (classification, regression, ensemble methods) in Python with TensorFlow and Scikit-Learn; applied systematic hyperparameter tuning, model evaluation, and performance optimisation.
- Built a Heart Disease Prediction system using Random Forest and Gradient Boosting ensembles — implemented modular Python code across data preprocessing, feature engineering, and evaluation stages with full technical documentation.

### AI Intern

Jul 2026 – Aug 2026

Techjays · Remote · 6 weeks

- Completed a structured AI Internship at Techjays (01 July 2026 – 13 August 2026), gaining practical, hands-on exposure to AI engineering practices and applied AI workflows.

## PROJECTS

### Agentic Data Pipeline Reliability Platform

Jun 2026 – Present

Python · CrewAI · Google Gemini 2.5 Flash · ChromaDB · FastAPI · Docker | [github.com/Veneel77](https://github.com/Veneel77)

- Built LLM-powered AI agents using CrewAI — implemented prompt logic, tool-calling interfaces, and structured outputs per design patterns; automated ETL incident lifecycle management covering failure classification, root cause analysis, fix recommendation, and report generation.
- Integrated agents with FastAPI REST backend and tool-calling interfaces; containerised with Docker Compose for reproducible deployment; implemented structured logging and observability hooks for agent health monitoring.
- Developed the RAG pipeline — document ingestion, chunking strategies, HuggingFace embedding generation, and ChromaDB vector search; wrote agent evaluation test cases and contributed to the evaluation harness measuring retrieval and response quality.

### RAG-Based Knowledge Assistant

Nov – Dec 2025

Python · LangChain · ChromaDB · HuggingFace Embeddings · FastAPI · Streamlit | [github.com/Veneel77/RAG-based-Knowledge-assistant](https://github.com/Veneel77/RAG-based-Knowledge-assistant)

- Designed and built an end-to-end RAG pipeline — document ingestion, chunking strategies, embedding generation, and vector search retrieval; integrated with a modular FastAPI backend and tool-calling interfaces with clean separation of concerns.
- Integrated HuggingFace embeddings with ChromaDB vector search; wrote agent evaluation test cases and participated in prompt review cycles; implemented unit tests; open-sourced under MIT licence with full documentation.

### Customer Churn Prediction System

Sep 2025

Python · Scikit-Learn · XGBoost · Pandas · Power BI | [github.com/Veneel77/Customer-Churn-Prediction-System](https://github.com/Veneel77/Customer-Churn-Prediction-System)

- Built a full ML pipeline on IBM Telco dataset (7,000+ records, 21 features) — feature engineering, ensemble model training with XGBoost (~85% accuracy); authored reproducible training pipeline artefacts and technical documentation.
- Delivered a centralised Power BI dashboard tracking model metrics, resource utilisation, and business impact for stakeholders.

## RESEARCH & PUBLICATIONS

### Alzheimer's Diagnosis via Deep Learning

JETIR, 2025

- Published peer-reviewed research applying CNN architectures and transfer learning (VGG16, ResNet50) on MRI medical imaging datasets for multi-class Alzheimer's stage classification — achieving >90% accuracy through hyperparameter tuning and data augmentation. Journal of Emerging Technologies and Innovative Research (JETIR), 2025.

## EDUCATION & ADDITIONAL INFO

### KSSEM, Bangalore — B.E. in AI & Data Science

Graduation: July 2025

Relevant Coursework: Deep Learning, Computer Vision, Machine Learning, Cloud Computing, Image Processing, Data Structures & Algorithms, NLP, DBMS

Certifications: Google Advanced Data Analytics · Google Cloud Kubernetes · Oracle GenAI Professional · Microsoft Azure AI-900

Honors & Awards: 1st Prize — AI Hackathon (College Level) · Solved 210+ Data Structures & Algorithms problems on LeetCode across arrays, trees, graphs, dynamic programming, and system design patterns

Interests: LLM & Agentic AI Research · RAG Systems & Prompt Engineering · Open Source Contribution · Cloud-Native AI Systems